

Problem Sheet 1

Instructions. For Questions 1–15 (1 mark each), shade the letter of the single best answer on the separate multiple-choice answer sheet. For Questions 16–20 (5 marks each), write your answers on A4 sheets and attach them to your answer sheet. Write your name and section on each A4 sheet and label each answer with its question number and subpart, where applicable. Show calculations and explain your reasoning in complete sentences. Include units where appropriate. Answer only what each question asks; do not add extra proofs, calculations, or examples. All datasets and laboratory models below are hypothetical.

1. A student observes that seeds in six moist trays germinate faster than seeds in six dry trays, then suggests that moisture generally speeds up germination. What kind of reasoning leads from these observations to the broader claim?
 - A. Deductive reasoning.
 - B. Inductive reasoning.
 - C. Proof by contradiction.
 - D. Mathematical induction.
 - E. None of A–D.
 - F. All of A–D.
2. Assume that every specimen in a collection is preserved in alcohol and that specimen S belongs to this collection. Which conclusion necessarily follows from these premises?
 - A. Every specimen preserved in alcohol belongs to this collection.
 - B. Specimen S is the only specimen preserved in alcohol.
 - C. Every biological specimen is preserved in alcohol.
 - D. Specimen S is preserved in alcohol.
 - E. None of A–D.
 - F. All of A–D.
3. A student checks a statement for the first 100 positive integers and finds that it holds each time. What has the student established?
 - A. Evidence for the statement, but not a proof for all positive integers.
 - B. A proof for every positive integer.
 - C. A counterexample to the statement.
 - D. The inductive step for every positive integer.
 - E. None of A–D.
 - F. All of A–D.
4. Which value is a counterexample to the claim “For every real number $x > 0$, we have $x^2 > x$ ”?
 - A. $x = 2$.
 - B. $x = 3$.
 - C. $x = \frac{1}{2}$.
 - D. $x = -1$.
 - E. None of A–D.
 - F. All of A–D.
5. A student wants to prove that the sum of any two even integers is even, but represents both integers by $2k$. Why is this insufficient?
 - A. Even integers cannot be written as $2k$.
 - B. It only covers pairs in which the two integers are equal.
 - C. It assumes that every integer is even.
 - D. It only covers negative integers.
 - E. None of A–D.
 - F. All of A–D.
6. A survey aims to describe the sleep duration of all first-year BS Biology students at a university. Forty of these students answer the survey. What is the population of interest?
 - A. The forty respondents.
 - B. All students at all universities.
 - C. Only first-year BS Biology students who sleep less than six hours.
 - D. All first-year BS Biology students at that university.
 - E. None of A–D.
 - F. All of A–D.
7. A dataset uses the code 1 for untreated seedlings and 2 for treated seedlings. What type of variable is treatment group?
 - A. Categorical, nominal.
 - B. Numerical, continuous.
 - C. Numerical, discrete.
 - D. Categorical, ordinal.
 - E. None of A–D.
 - F. All of A–D.
8. A researcher estimates the average height of all seedlings in a nursery by measuring only seedlings on its sunniest shelf. What concern remains even if many seedlings on that shelf are measured?
 - A. A large sample must have a mean of zero.
 - B. Height becomes a categorical variable.
 - C. The selected shelf may not represent the entire nursery.
 - D. Measuring more seedlings always removes selection bias.
 - E. None of A–D.
 - F. All of A–D.

9. Which information belongs in a data dictionary for seedling height?
- What the height variable measures and when it is measured.
 - The unit used for recorded heights.
 - The allowed range of values.
 - The code for an unavailable value and what that code means.
 - None of A–D.
 - All of A–D.
10. Which graph is most suitable for comparing the numbers of specimens in four different species categories?
- A histogram of numerical intervals.
 - A bar chart.
 - A scatterplot of two numerical measurements.
 - A line graph of change over time.
 - None of A–D.
 - All of A–D.
11. A student records both leaf length and leaf width, in centimetres, for each of 30 leaves. Which display is most suitable for investigating the relationship between these two measurements?
- A frequency table of species names.
 - A pie chart of leaf identifiers.
 - A bar chart of the total number of leaves.
 - A scatterplot of width against length.
 - None of A–D.
 - All of A–D.
12. In Tray A, 90% of 20 seeds germinate. In Tray B, 50% of 60 seeds germinate. What percentage of all 80 seeds germinates?
- 50%.
 - 65%.
 - 70%.
 - 90%.
 - None of A–D.
 - All of A–D.
13. A study records seedling heights in centimetres. In what units should the variance and standard deviation be reported, respectively?
- Centimetres and square centimetres.
 - Centimetres and centimetres.
 - Square centimetres and centimetres.
 - Square centimetres and square centimetres.
 - None of A–D.
 - All of A–D.
14. A dataset has $Q_1 = 4$ cm and $Q_3 = 8$ cm. Under the 1.5 IQR rule, which recorded height is flagged as a potential outlier?
- 15 cm.
 - 14 cm.
 - 8 cm.
 - 4 cm.
 - None of A–D.
 - All of A–D.
15. A valid height measurement is flagged as a potential outlier during analysis. What is the appropriate next action?
- Delete it solely because it is beyond an outlier fence.
 - Investigate its context and document any decision to retain or exclude it.
 - Replace it silently with the mean.
 - Treat the flag alone as proof that the measurement was fabricated.
 - None of A–D.
 - All of A–D.

For Questions 16–18, state the relevant assumptions, justify each step, and end with a conclusion. An integer is even if it equals $2r$ for some integer r , and odd if it equals $2r + 1$ for some integer r . You may use the fact that sums and differences of integers are integers.

16. Prove that the difference of any two odd integers is even. Your proof must cover arbitrary odd integers, including when their difference is zero or negative. Use separate integer variables for the two odd integers.
17. Prove that there is no largest even integer. Begin by assuming that a largest even integer M exists. Construct an even integer larger than M and explain why this contradicts the assumption.
18. An idealized cell model starts with three cells at stage 0. At each next stage, every cell is replaced by exactly two daughter cells, with no other additions or losses. Thus,

$$C_0 = 3, \quad C_{n+1} = 2C_n \quad \text{for every integer } n \geq 0.$$

Prove by mathematical induction that $C_n = 3 \cdot 2^n$ for every integer $n \geq 0$ by answering the following parts.

- (a) Identify the starting value of n and verify that the formula holds at that stage. **(1 mark)**
 - (b) State your assumption for an arbitrary integer $k \geq 0$. **(1 mark)**
 - (c) Use the model rule and your inductive hypothesis to prove the formula for stage $k + 1$. Then state your conclusion by mathematical induction. **(3 marks)**
19. A class plants 50 seeds and records one row per seed. The columns are seed_id, germinated (yes/no), and height14_cm (seedling height 14 days after planting, in centimetres).
 - (a) State what one observation is and identify the column that uniquely labels it. **(1 mark)**
 - (b) Classify germinated and height14_cm by variable type. **(2 marks)**
 - (c) Seed S17 does not germinate. Explain why entering 0 in its height cell is misleading. Describe how to record the unavailable height and its reason. **(2 marks)**
 20. Five seeds have the following germination times, in days:

$$3, \quad 4, \quad 4, \quad 5, \quad 9.$$
 - (a) Calculate the mean and median germination times. **(2 marks)**
 - (b) State the mode and calculate the range. **(2 marks)**
 - (c) If the final value were 19 days instead of 9 days, which of the mean and median would remain unchanged? Explain briefly. **(1 mark)**

End of questionnaire – 40 marks